

Mirazul Islam, PhD

Email: md.mirazul.islam@vanderbilt.edu; mirazul.du.bc@gmail.com

Website: <https://mirazul302.github.io/>

Cancer biologist specializing in cancer evolution, intestinal pathophysiology, and single-cell genomics. My research integrates experimental and computational approaches to develop lineage-tracing and CRISPR technologies, define mechanisms of tissue regeneration and cellular quiescence, and uncover the genetic and microenvironmental processes that drive precancer-to-cancer progression. My work has been published in **Nature (2024)** and **Cell (2026)**, with additional manuscripts under revision in **Cell** and **Nature Biotechnology**. I have received independent research funding as PI, including a Career Enhancement Award, VICC Pilot Project Award, and DDRC Pilot Project Award, and received an **NCI Early K99/R00** impact score of 26 (8th percentile).

Research Interests

Cancer evolution and precancer-to-cancer progression; intestinal stem cell and developmental biology; tissue regeneration and homeostasis; cancer genomics and single-cell multi-omics; lineage tracing and molecular recording; extracellular vesicles (EV)-mediated live cell-cell communication tracing

Education and Training

- **Vanderbilt University Medical Center** - Nashville, TN
NIH T32 Postdoctoral Fellow (Oct 2024 - Present)
- **Vanderbilt University** - Nashville, TN
PhD in Cell and Developmental Biology (Aug 2019–Sept 2024)
Dissertation: Recording mammalian development, regeneration, and tumorigenesis at single-cell resolution.
Dissertation committee: Drs. James Goldenring, Ian Macara, Christopher Wright, and Omer Yilmaz
- **University of Michigan** - Ann Arbor, MI
Graduate Student in Pathology (July 2018 - July 2019)
- **University of Malaya** - Kuala Lumpur, Malaysia
MS in Medical Science (Sept 2014 - June 2016)
- **University of Dhaka** - Dhaka, Bangladesh
BS in Biochemistry and Molecular Biology (Jan 2010 - Mar 2014)

Academic Research Experience

Vanderbilt University Medical Center - Nashville, TN

NIH T32 Postdoctoral Fellow, Robert Coffey's Lab, Oct 2024 - Present

- Identified and validated a previously unrecognized intestinal stem cell population (pISCs) in the mouse intestine
- Discovered a new regulatory mechanism of cellular quiescence in the intestinal epithelium
- Defined clonal dynamics of the human and mouse intestinal epithelium during homeostasis and regeneration

Vanderbilt University - Nashville, TN

PhD in Cell and Developmental Biology, Ken Lau's Lab, Aug 2019 - Sept 2024

- Developed NSC-seq, a single-cell multimodal platform, for *in vivo* lineage tracing and molecular recording
- Designed a CRISPR-based molecular clock to record the timing of cellular events *in vivo*

- Revealed the clonal origins of colonic precancers and their evolutionary progression to malignancy

Broad Institute of MIT and Harvard & Dana-Farber Cancer Institute - Cambridge, MA

Research Specialist, Adam Bass's Lab, July 2016 - June 2018

- Analyzed genomic alterations in cancer patients as part of The Cancer Genome Atlas (TCGA)
- Performed gene-expression profiling to characterize esophageal stem/progenitor cells
- Identified WRN as a synthetic-lethal target in microsatellite instability (MSI) cancers using CRISPR screening

Other Research Experiences:

- **University of Michigan** - Generated CRISPR-barcoded cell lines for lineage tracking
- **German Cancer Research Center** (Christoph Plass's Lab) - Developed methods for integrating DNA methylation and gene expression data
- **University of Malaya** (Zahurin Mohamed's Lab) - Studied cytokine expression profiles in AML patients
- **Children's Hospital Los Angeles** (Paul Gaynon's Lab) - Conducted retrospective analyses of AML treatment outcomes
- **Lawrence Berkeley National Laboratory** (Priscilla Cooper's Lab) - Applied CRISPR-Cas9 for targeted gene knockout
- **Academia Sinica, Taiwan** (Sue Lin-Chao's Lab) - Investigated the role of the MLL-GAS7 oncogenic fusion protein in AML

Publications

Research Articles:

1. **Islam M**, Yang Y, Simmons AJ, Shah VM, Pavan MK, Xu Y, Tasneem N, Chen Z, Trinh LT, Molina P, Ramirez-Solano MA, Sadien I, Dou J, Chen K, Magnuson MA, Rathmell J, Macara IG, Winton DJ, Liu Q, Zafar H, Kalhor R, Church GM, Shrubsole MJ, Coffey RJ#, Lau KS#. *Temporal recording of mammalian development and precancer. **Nature***, 634, 1187–1195, 2024
2. **Islam M#**, Bechard M, Yang Y, Simmons AJ, Dagotto CE, Xu Y, Aramandla R, Higginbotham J, Zhao P, Cao Z, Smith RT, Tasneem N, Glass S, Markham N, Revetta R, Ramirez M, Liu Q, Franklin J, Lau KS#, Coffey RJ#. *Unbiased recording of clonal potency reveals species-specific regulation of mouse intestine. **Cell** 2026, in press.*
3. **Islam M#**, Yang Y, Simmons AJ, Bechard M, Dagotto CE, Xu Y, Aramandla R, Cao Z, Revetta R, Franklin J, Lau KS, Coffey RJ#. *Clonal architecture of human colonic epithelium. **Cell** 2026, under revision.*
4. **Islam M#**, Yang Y, Simmons AJ, Xu Y, Fisher EL, Deng W, Grieb BC, Molina P, Caestecker CD, Ramirez-Solano MA, Liu Q, Tansey WP, Macara IG, Rathmell JC, Coffey RJ, Lau KS#. *Scalable single-cell pooled CRISPR screens with knockout vector libraries. **Nature Biotechnology** 2025, under revision.*
5. Duronio GN, Liang X, Hebbar P, **Islam M**, Spisak, S, Sethi, N#. Truncating SOX9 alterations are heterozygous null alleles in genome stable colorectal cancer. ***Gastro Hep Advances***, 1(5), 709-713, 2022

6. Chen B*, Scurrah C*, McKinley ET, Simmons AJ, Ramirez-Solano MA, Zhu X, Markham NO, Heiser CN, Vega PN, Rolong A, Kim H, Sheng Q, Drewes JL, Zhou Y, Southard-Smith AN, Xu Y, Ro J, Jones AL, Revetta F, Berry LD, Hiroaki Niitsu, **Islam M**, Pelka K, Hofree M, Chen JH, Sarkizova S, Ng K, Giannakis M, Boland GM, Aguirre AJ, Anderson AC, Rozenblatt-Rosen O, Regev A, Hacohen N, Kawasaki K, Sato T, Goettel JA, Grady WM, Zheng W, Washington MK, Cai Q, Sears CL, Goldenring JR, Franklin JL, Su T, Huh WJ, Vandekar S, Roland JT, Liu Q, Coffey JC#, Shrubsole JM#, Lau KS#. Differential pre-malignant programs and microenvironment chart distinct paths to malignancy in human colorectal polyps. **Cell**. 184 (26), 6262-6280, 2021
7. Tang Q, Efe G, Chiarella AM, Leung J, Chen M, Yamazoe T, Su Z, Pitarresi JR, Li J, **Islam M**, Karakasheva T, Klein-Szanto AJ, Pan S, Hu J, Natsugoe S, Gu W, Stanger BZ, Wong KK, Diehl JA, Bass AJ, Nakagawa H, Murphy ME, Rustgi AK#. Mutant p53 regulates Survivin to foster lung metastasis. **Genes & development**. 35 (7-8), 2021
8. Tang Q, Lento A, Suzuki K, Efe G, Karakasheva T, Long A, Giroux V, **Islam M**, Wileyto EP, Klein-Szanto AJ, Nakagawa H, Bass AJ, Rustgi AK#. Rab11-FIP1 mediates epithelial-mesenchymal transition and invasion in esophageal cancer. **EMBO reports**. 22(2), 2021
9. Sethi N*, Kikuchi O*, McFarland J, Zhang Y, Chung M, Kafker N, **Islam M**, Chakraborty A, Kaelin WG, Bass AJ#. Mutant p53 Induces a Hypoxia Transcriptional Program in Gastric and Esophageal Adenocarcinoma. **JCI Insight**. 4 (15), August 2019
10. Chan E M*, Shibue T*, McFarland JM, Gaeta B, Ghandi M, Dumont N, Gonzalez A, McPartlan JS, Li T, Zhang Y, Liu JN, Lazaro JB, Gu P, Pieltz CG, Apffel A, Ali SO, Deasy R, Keskula P, Ng RW, Roberts EA, Reznichenko E, Leung L, Alimova M, Schenone M, **Islam M**, Maruvka YE, Liu Y, Roper J, Raghavan S, Giannakis M, Tseng YY, Nagel ZD, D'Andrea A, Root DE, Boehm JS, Getz G, Chang S, Golub TR, Tsherniak A, Vazquez F#, Bass AJ#. WRN Helicase is a Synthetic Lethal Target in Microsatellite Unstable Cancers. **Nature**. 568(7753), April 2019
11. Korkut A, Zaidi S, Kanchi RS,...**Islam M**,....The Cancer Genome Atlas Research Network, Mishra L#, Akbani R#. A Pan-Cancer Analysis Reveals High-Frequency Genetic Alterations in Mediators of Signaling by the TGF- β Superfamily. **Cell Systems**. 7(4), October 2018
12. McBrayer SK, Mayers JR, DiNatale GJ, Shi DD, Khanal J, Chakraborty AA, Sarosiek KA, Briggs KJ, Robbins AK, Sewastianik T, Shareef SJ, Olenchok BA, Parker SJ, Tateishi K, Spinelli JB, **Islam M**, Haigis MC, Looper RE, Ligon KL, Bernstein BE, Carrasco RD, Cahill DP, Asara JM, Metallo CM, Yennawar NH, Heiden MV, Kaelin WG#. Transaminase inhibition by 2-hydroxyglutarate impairs glutamate biosynthesis and redox homeostasis in glioma. **Cell**. 175(1), September 2018
13. Liu Y*, Sethi NS*, Hinoue T*, Schneider BG*, Cherniack AD, Vega FS, Seoane JA, Bowlby R, **Islam M**,..., Thorsson V#, Bass AJ#, Laird PW#. Comparative Molecular Analysis of Gastrointestinal Adenocarcinomas. **Cancer Cell**. 33(4), April 2018
14. Pectasides E*, Stachler MD*, Derks S*, Liu Y*, Maron S*, **Islam M**,..... Bass AJ#, Catenacci DV#. Genomic Heterogeneity as a Barrier to Precision Medicine in Gastroesophageal Adenocarcinoma. **Cancer Discovery**. 7(11), November 2017
15. **Islam M**#, Mohamed EH, Esa E, Kamaluddin NR, Zain SM, Yusoff YM, Assenov Y, Mohamed Z#, Zakaria Z. Circulating cytokines and small molecules follow distinct expression patterns in acute myeloid leukemia. **British Journal of Cancer**. 117(10), September 2017
16. Giroux V, Lento AA, **Islam M**, Pitarresi JR, Kharbanda A, Hamilton KE, Whelan KA, Long A, Rhoades B, Tang Q, Nakagawa H, Lengner CJ, Bass AJ, Wileyto EP, Klein-Szanto AJ, Wang TC, Rustgi AK#. Novel long-lived esophageal progenitor cells contribute to homeostasis and regeneration. **Journal of Clinical Investigation**. 127(6), June 2017

17. **Islam M**, Mohamed Z#, and Assenov Y#. *Differential Analysis of Genetic, Epigenetic, and Cytogenetic Abnormalities in AML. **International Journal of Genomics***. 2913648, June 2017
18. **Islam M#**, Akataruzzaman M, and Mahmud Z#. *Comparative Sequence-Structure Analysis of Aves Insulin. **Bioinformation***. 11(2), February 2015
19. Aktaruzzaman M, **Islam M**, Mohamed Z, Islam MS, Howlader MMR. Therapeutic efficacy of ivermectin, fenbendazole and albendazole against naturally occurring gastrointestinal nematodiasis infection in Black Bengal Goat of Bangladesh. ***International Journal of Biological Research***. 3 (1), 42-45. 2015
20. Haque ME, Sultana A, Shibib BA, **Islam M**. Antimicrobial, antioxidant and cytotoxic activities of *Callistemon citrinus* (Curtis) Skeels. ***Dhaka Univ J Pharm Sci***. 11 (1), 51-4. 2012

Review Articles:

1. **Islam M**, Chen B, Spraggins JM, Kelly RT, Lau KS#. *Use of Single Cell-omic Technologies to Study the Gastrointestinal Tract and Diseases, From Single Cell Identities to Patient Features. **Gastroenterology***, 159 (2), 453-466. 2020
2. **Islam M#**, Mohamed Z#. *Computational and pharmacological target of neurovascular unit for drug design and delivery. **BioMed Research International***. 2015(731292), Oct 2015
3. **Islam M#**. *Role of bioinformatics in developing country: Bangladesh. **Current Trends in Technology & Science***. 2 (1), 160-165. 2013
4. **Islam M**, Kabir Y. *Electronic Health Record Database for Modernizing Healthcare and Medical Research in the Perspective of Bangladesh. **Bang J of Med Sci***. 17 (2), 171-175. 2011

Research Grants & Fellowships

- **SPORE Career Enhancement Award (\$75,000) - 2026, PI**
- **VICC Pilot Project Award (\$40,000) - 2026, PI**
- **DDRC Pilot Project Award (\$25,000) - 2026, PI**
- **DDRC Core Complementary Award (\$2,500) - 2026, PI**
- **NCI Early K99/R00 - Identifying the genetic and microenvironmental bottlenecks of colorectal precancer-to-cancer progression at single-cell resolution; Impact Score: 26 (A0: 8th percentile; A1: under review)**
- **NIH T32 Postdoctoral Training Fellowship (T32DK007673) – 2025 - Present**
- Early-Onset CRC Foundation Award - Under review
- VICTR Award - Under review
- Vanderbilt DDRC Pilot Project Award - 2023, Co-investigator/trainee with Ken Lau
- GI SPORE and HTAN Funding Support - Co-investigator/trainee with Ken Lau and Robert Coffey

Awards & Honors

- Provost Pathbreaking Discovery Award - Vanderbilt University, 2024
- VICC Graduate Student of the Year Award - Vanderbilt University, 2024
- Graduate Travel Fellowship - Vanderbilt University, 2023
- Graduate Student Fellowship - University of Michigan, 2018

- Travel Grant for 13th APFP Meeting, Thailand, Bangkok, 2016
- TIGP International Internship Scholarship - Academia Sinica, Taiwan, 2015
- Summer Oncology Fellowship - CHLA, USA, 2015

Invited Talks and Seminars:

1. Recording cellular life histories: multimodal insights into colorectal cancer. University of Chicago, Department of Medicine Grand Round, May 2026.
2. pISCs: a new population of intestinal stem cells. NCI Program on the Origins of Gastroesophageal Cancers Annual Meeting 2026, Nashville, TN, March 3-4, 2026.
3. Recording cellular life histories: multimodal insights into development and cancer. University of Alabama, Birmingham, March 2026.
4. Recording cellular life histories: multimodal insights into development and cancer. University of Oklahoma, Oklahoma, Dec 2025.
5. Quiescent stem cells as a defining feature of the intestinal hierarchy. FASEB GI Meeting. Southbridge, Massachusetts, September 2025.
6. Recording clonal dynamics of mammalian intestinal epithelium. EMBO Workshop: Lineage tracing: Dynamics, cellular memory, and somatic evolution. Spain, September 2025.
7. Recording cellular life histories: multimodal insights into development and cancer. University of Maryland, Baltimore, December 2025.
8. Recording cellular life histories: multimodal insights into development and cancer. University of Cincinnati, Ohio, November 2025.
9. A molecular clock reveals evolutionary dynamics in mammalian intestine and tumorigenesis. Moffitt Cancer Center, Florida, October 2025.
10. Species-specific regulation of mouse and human intestinal stem cells. The Epithelial Biology Center Symposium, VUMC, Nashville, March 2025
11. Clonal dynamics of mammalian intestinal epithelium and tumorigenesis. Broad Institute of MIT and Harvard, January 2025
12. Temporal recording of mammalian development and regeneration. Genome Engineering Seminar Series at Harvard Medical School, December 2024
13. Exploring clonal dynamics in intestinal epithelial regeneration and early cancer. BME Seminar, Johns Hopkins University, September 2024
14. Clonal dynamics of homeostatic and regenerative intestinal epithelium. Rockefeller University, September 2024
15. Exploring clonal dynamics in intestinal epithelial regeneration and early cancer. Systems Biology Seminar, MD Anderson Cancer Center, August 2024
16. Single-cell recording reveals the polyclonal origins of colonic precancers. Vanderbilt-Ingram Cancer Center, Annual retreat, 2024
17. Identifying an embryonic revival stem cell population in the intestine through temporal recording of development. Vanderbilt Center for Stem Cell Biology (VCSCB) Symposium, Nashville, 2024
18. Clonal analysis of intestinal epithelium following irradiation. Epithelial Biology Center, Nashville, 2023
19. Clonal analysis of intestinal tumor supports an oligoclonal model of tumor formation. Program in Developmental Biology, annual retreat, Vanderbilt, Nashville, 2022
20. Direct sgRNA capture enable whole organism lineage tracking and CRISPR screen at single cell resolution. Cell and Developmental Biology, annual retreat, Vanderbilt, Nashville, 2022

Conference Proceedings:

1. Islam M, et al. Identifying an embryonic revival stem cell population in the intestine through developmental history recording. 2nd Gastrointestinal Conference. Cancun, **Mexico**, 2024
2. Islam M, et al. Temporal recording of mammalian development and precancer. HTAN and HuBMAP Joint Meeting. Stanford, **CA**, 2024
3. Islam M, et al. Temporal recording of mammalian development and cancer. Single Cell Biology: From Development to Cancer. Keystone, **CO**, 2023
4. Islam M. et al. Native sgRNA capture and sequencing (NSC-seq) reveals tumor formation in Apc (Min/+) mice and in human colorectal cancer is an oligoclonal process. AACR Special Conference on Colorectal Cancer. Portland, **Oregon**, 2022
5. Chen B, Scurrah C, Mckinley E, Islam M, et al. Human Colorectal pre-cancer atlas identifies distinct molecular programs underlying to major subclasses of pre-malignant tumors. Stem Cell and Cancer. IRB Barcelona, **Spain**, 2021
6. Islam M. et al. Circulating cytokines, chemokines, and small molecules follow distinct expression patterns in acute myeloid leukemia. Tumor Immunology and Oncology. **Boston**. 2017.
7. Islam M, Mohamed Z, and Assenov Y. Differential analysis of genetic, epigenetic, and cytogenetic abnormalities in AML. The 13th Asia Pacific Federation of Pharmacologist (APFP) Meeting, Bangkok, **Thailand**. February 2016

Academic Service & Professional Memberships

- **Editorial & Reviewing Activities:** *Blood*, *Blood Advances*, *Cancer Letters*, *BMC Cancer*, *BMC Genomics*, *International Journal of Genomics*, *Clinical Cancer Research*
- **Grant Reviewer:** MEET2WIN Oncology Partnering Grant, MATWIN, France, 2018
- **Professional Memberships:** AACR, GSA, AAAS, AHA

Mentorship & Teaching

- **Mentored Undergraduate/High School Students:** Vishal M. Shah (PhD Student, Boston University); Jiawei Wang (MD Student, Australia); Kyla Johnson (Undergrad, Fisk University); Dharani Bangaru (Nolensville High School)
- **Mentored Graduate Students:** Carina Dagotto (PhD Student); Pushpita Biswas (incoming PhD Student). Assisted numerous students to run NSC-seq experiments and data analysis for lineage tracking and CRISPR screen
- **Teaching-Related Experience:** Biology of Cancer; Epithelial Biology (grad level)
- **Outreach:** Founder & President of *Voluntary Work for Social Change* (VWSC), an initiative focused on developing quality educational content for high school students
- **Teaching Interest:** Clonal dynamics of stem cells and development; Mechanism of tissue regeneration; Systems biology; Single-cell multi-omics; Data science

Referees:

- Dr. Ken Lau, Vanderbilt University
- Dr. Robert Coffey, Vanderbilt University Medical Center
- Dr. Anil Rustgi, Columbia University Medical Center
- Dr. Adam Bass, Memorial Sloan Kettering Cancer Center
- Dr. Omer Yilmaz, MIT